Entity Relationship Diagram

ERD (Entity Relationship Diagram)

- is modeling tool used to depict graphically a database design before it is actually implemented.
- It has three basic components, namely, an **Entity**, **Relationship** and an **Attribute**.

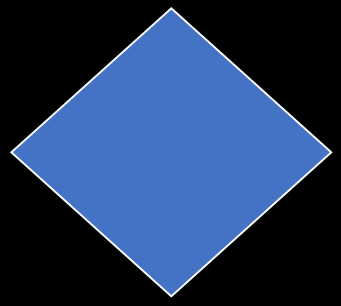
Entity

- is a real-world object distinguishable or unique from other objects.
- An entity can be a **concrete** or **physical** object like *employee*, *student*, *faculty*, *customer* etc. Or it could also be **conceptual** or **abstract** like *transaction*, *order*, *course*, *subjects* etc.
- It can be thought of as a noun like *student, employee* etc.
- It is normally represented by a rectangle shape.

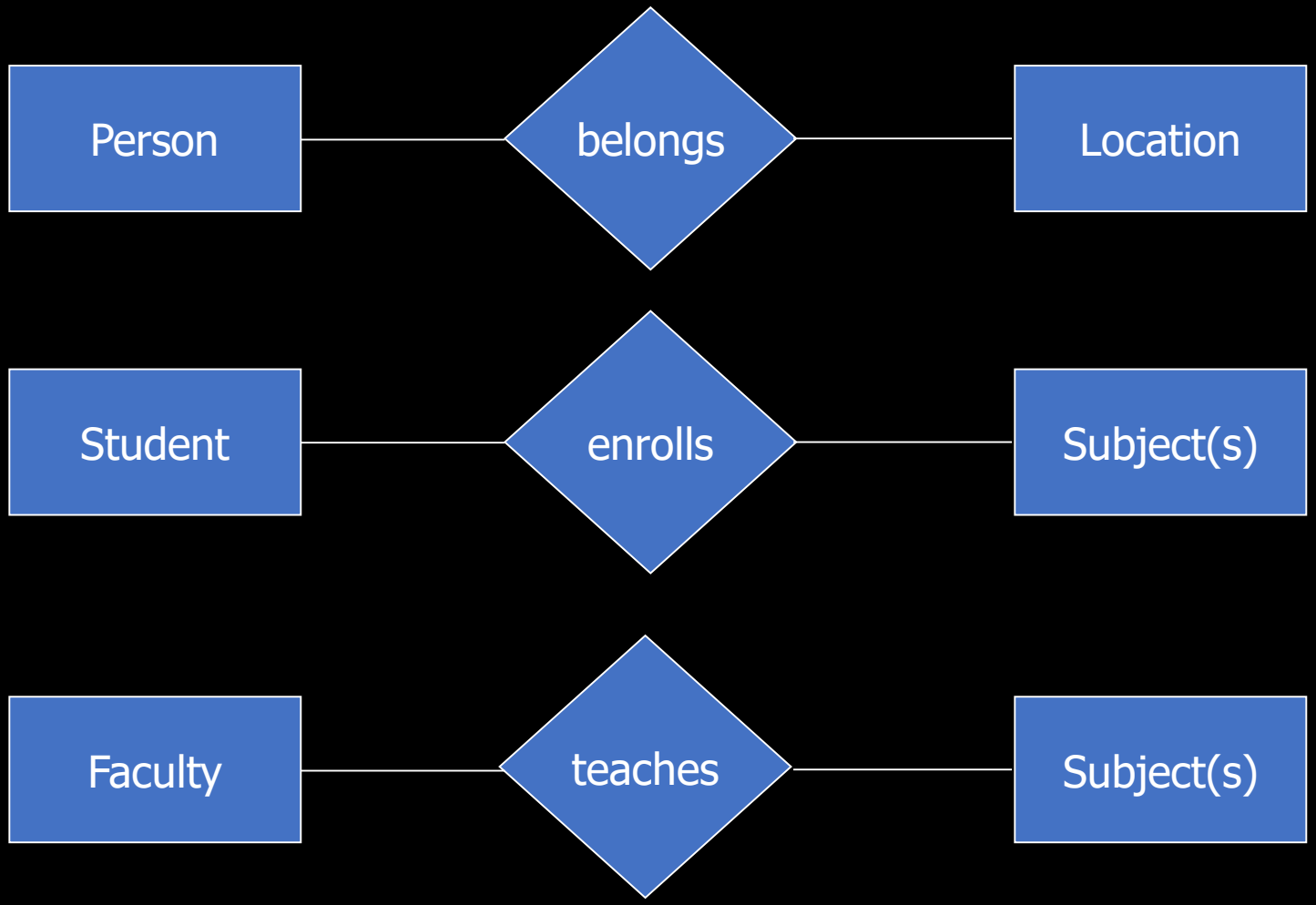


Relationship

- is a way of relating one entity to another. Entities can therefore participate in a relationship.
- it is commonly thought as a *verb* connecting the *entities* or *nouns.*
- It is normally represented by a diamond shape.



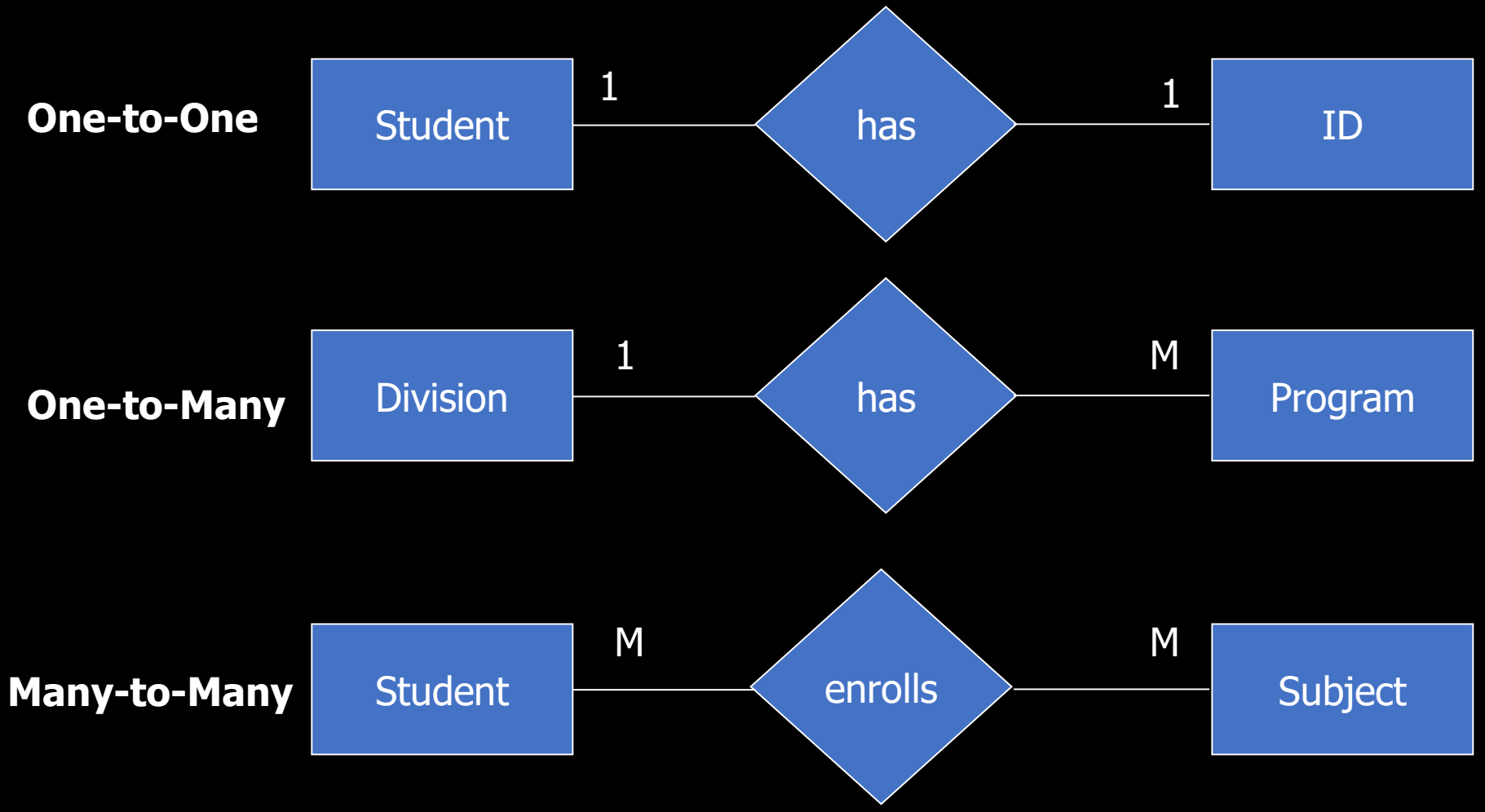
Example of Entities with Relationship



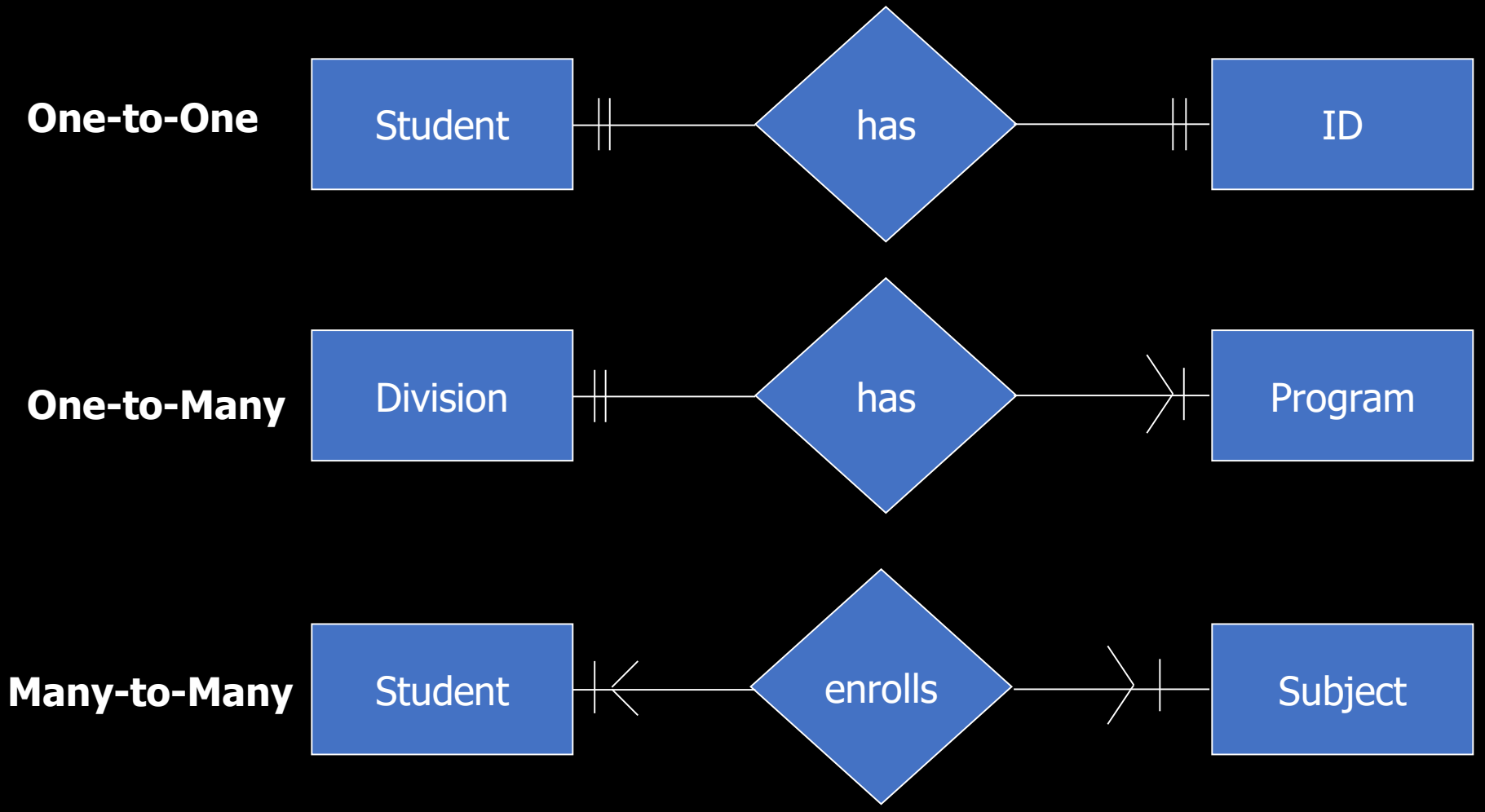
Types of Relationship

- One-to-One
- One-to-Many
- Many-to-Many

Types of Relationship



Types of Relationship (Crow Notation)

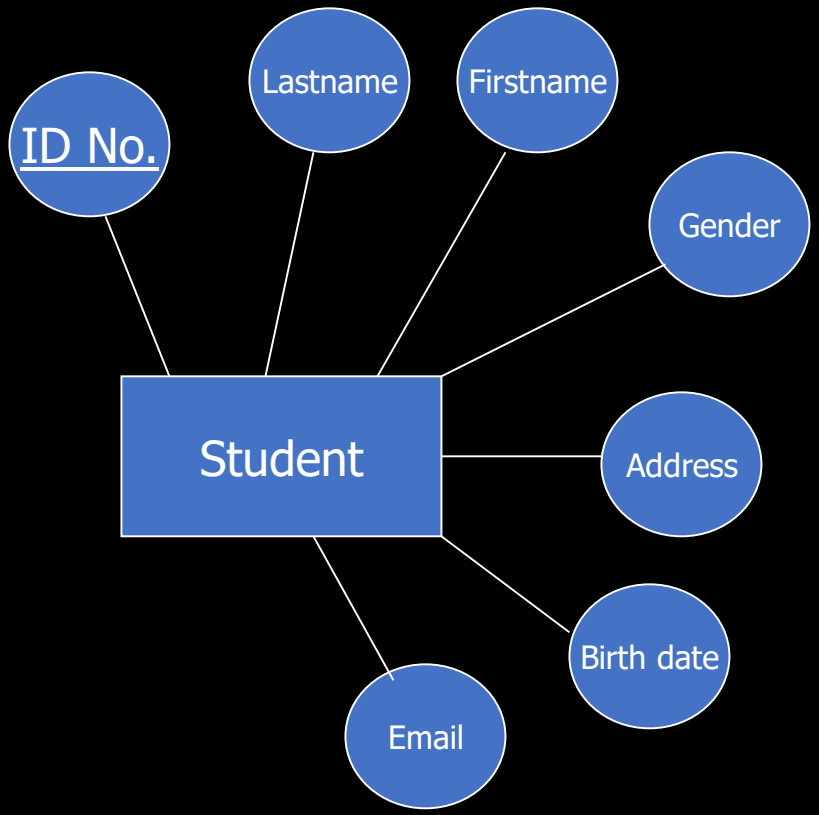


Attribute

- Refers to the characteristic or basic fact or field of an Entity or Relationship.
- For example a Student entity could have the following **attributes** *ID Number*, *Last Name, First Name, Address, Birth Date* etc.
- A relationship could also have an attribute for example an Entity name **Student** enrolls (relationship) to a **Course/Program**. Now, when you enroll you enroll on a certain date so you will have an attribute of *Enrollment Date* under **Enroll** relationship.
- It is normally represented by a **circle**.

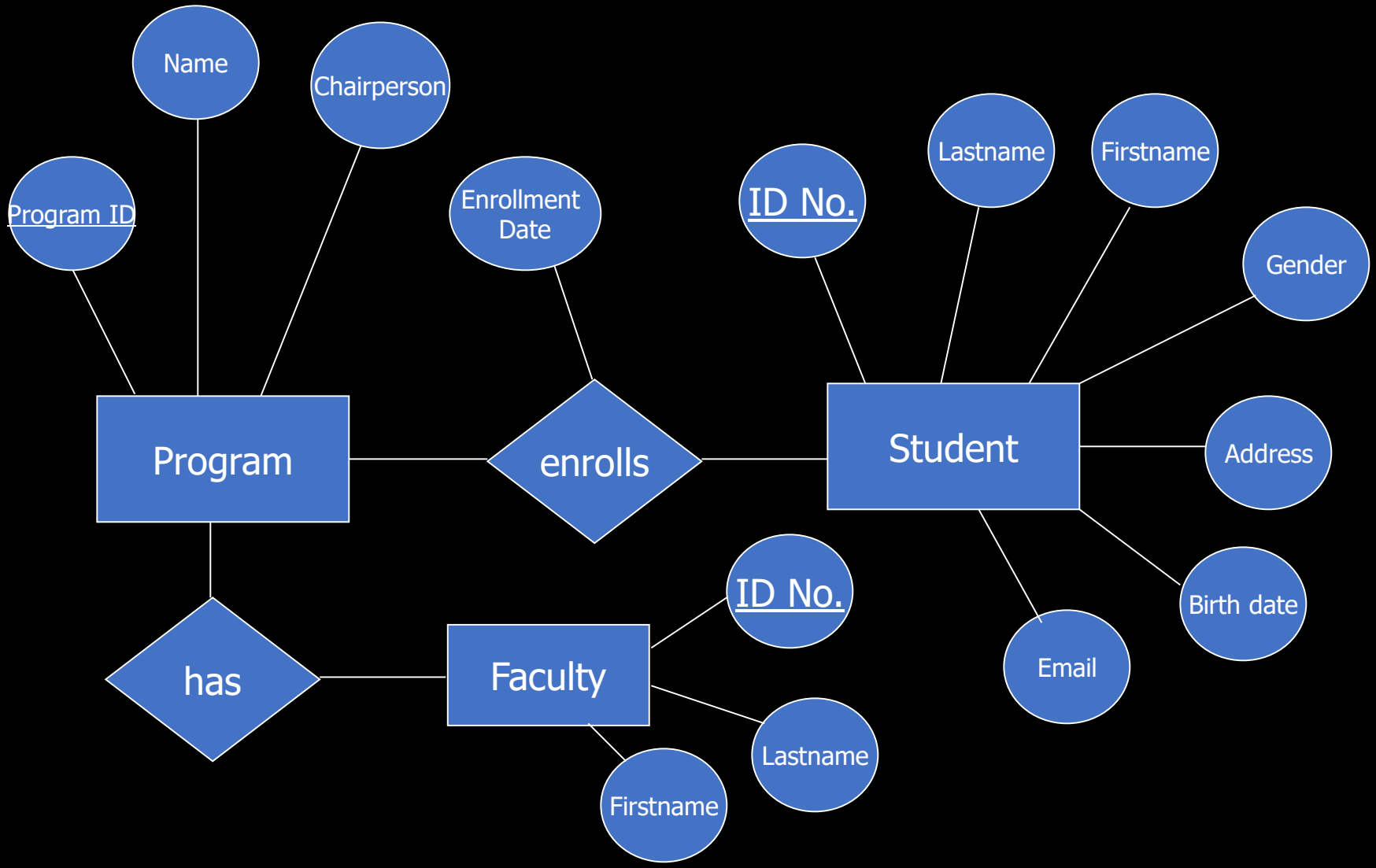


Example of Attributes

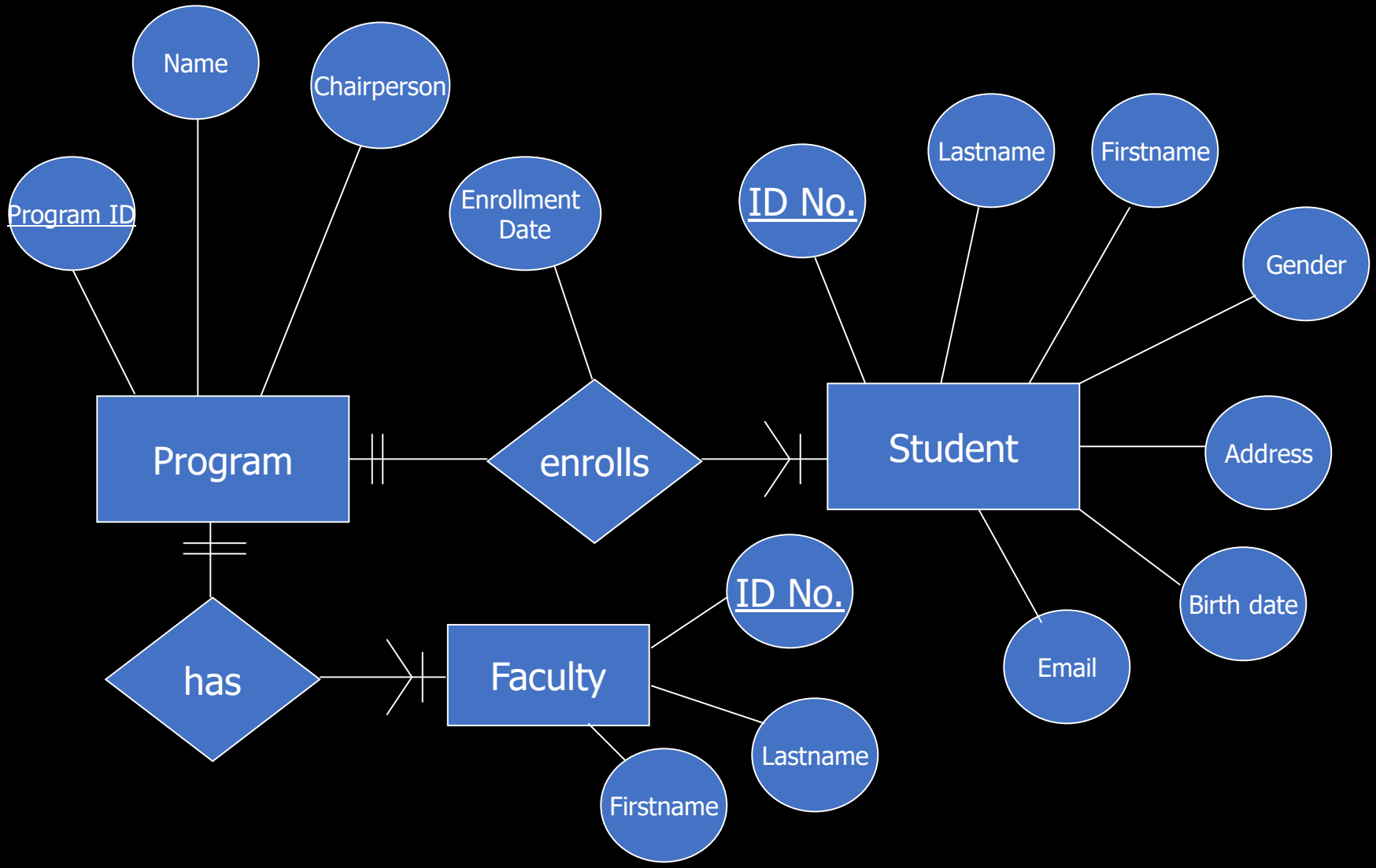


Note : A Primary Key is Underlined, in this case the ID No.

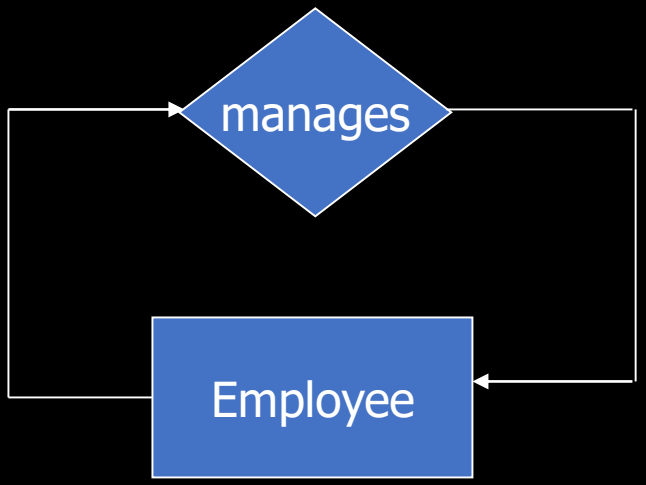
Example of Attributes



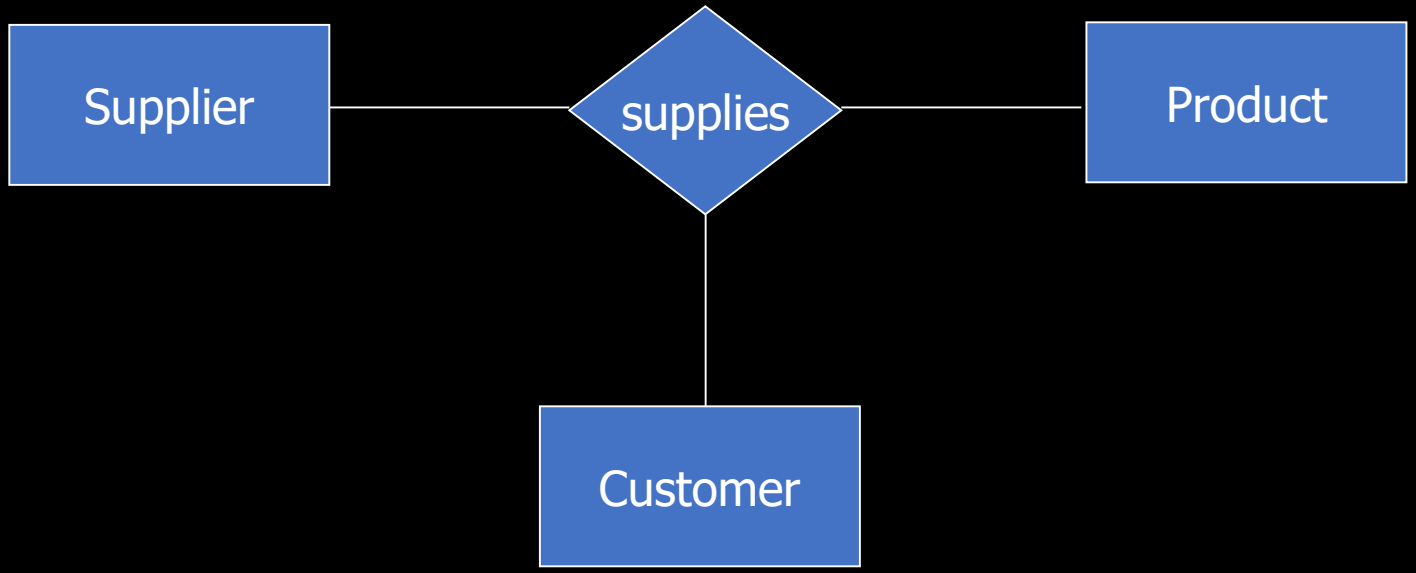
Example of Attributes



Recursive ERD



ERD with Ternary Relationship



ERD (More Examples)

